CSE 29 Week 3 Discussion

POINTERS, Terminal I/O, Project 2, Hashing, and more

Announcements

- Project 2 due **10PM** Friday, 10/23
- Project 1 resubmission due **10PM** Friday, 10/23
- Lab 3 check-off due **11AM** Monday, 10/19
- Review Question answers are being posted now.

Pointers review

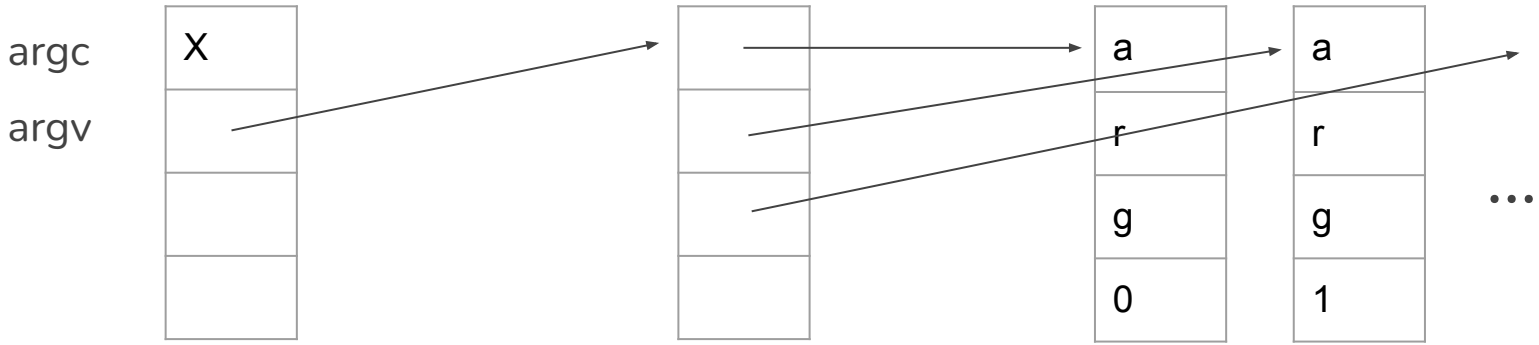
- char* var —> instantiate a pointer
- *var —> dereference a pointer
- &var —> view the memory address of the variable

argc, argv

```
// int main(int argc, char* argv[]){
int main(int argc, char** argv){

    return 0;
}
```

- argc = number of arguments passed into the program
- argv = array of strings containing the arguments



sizeof()

- Accepts a **data type** (e.g. int, char, char*, etc) or a **variable**
- Returns the **number of bytes in memory** that data type or variable takes

- Demo

Project 2 Introduction

PA2 - Hashing and Passwords: Due Friday 10/23 at 10:00pm

Cryptographic hash functions take an input of arbitrary length and produces a fixed length output. The special features are that the outputs are *deterministic* (the same input always generates the same output) and yet the outputs are apparently "random" – two similar inputs are likely to produce very different outputs, and it's difficult to determine the input by looking at the ouput.

A common application of hashing is in storing passwords. Many systems store only the hash of a user's password along with their username. This ensures that even if someone gains access to the stored data, users' actual passwords are not exposed. When a user types in a password on such a system, the password handling software grants access if the hash value generated from the user's entry matches the hash stored in the password database.

Questions

- **Project 2**
- Lecture content: pointers, hashing, ...
- Labs: unix commands, git, vim, ...
- etc